

MATH 161C Fall 2025 Project Reports

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Report 1: Logistic Regression - Predictors of Support for Same-Sex Marriage

Research Question and Context

GSS has consistently monitored the evolving attitudes toward same sex marriage since 1988 till 2024, and asks:

"Do you agree or disagree?"

Homosexual couples should have the right to marry one another, where responses are ranged on a scale of:

"Strongly Disagree," "Disagree," "Neutral," "Agree," "Strongly Agree," and "Can't Choose."

They are recorded here as binary for logistic regression, excluding the "Can't Choose" option:

H_0 : (agree/strongly agree) vs. H_a : (neutral/disagree/strongly disagree)

With the provided data set, the question that arises:

To what extent do age and political party affiliation predict support for same sex marriage among U.S. adults?

We believe the best choices to operate a logistic regression when considering the question are obviously age, political party, year, and the same marriage results. Throughout the years, society's ideology typically changes from time due various events and progression for acceptance. GSS displays the graphs of people agreeing and disagreeing about same-sex marriage. When bringing into account all variables, it is considered that around 45% of the population in 2022 strongly agreed for same-sex marriage. While combining strongly disagree, disagree, and neutral about 27% of the population voted for these options. Not to dismiss the other documented years of this survey, but it is safe to say that the agreeing options are increasing and the disagreeing options are decreasing.

Statistical Method and Assumptions

After using JMP to create a binary logistic regression model, the fitted result is:

$$\text{logist}(P(\text{support} = 1)) = \beta_0 + \beta_1(\text{AGE}) + \beta_2(\text{PARTYID}),$$

where AGE in years is a continuous predictor, and PARTYID is categorical predictor. Using the model selection option it was able to create a forward stepwise regression with minimum BIC.

Key assumptions include:

- (1) The response variable is binary (1 = supports same sex marriage, 0 = does not support)
- (2) Observations are independent of one another
- (3) Sufficient amount of sample size ($n \approx 75,000$)
- (4) No multicollinearity
- (5) Linear relationship

Justification

The research question is trying to understand how the effects of age and political party affiliation are associated with support of same sex marriage within our society throughout the years. The GSS was able to provide an abundant amount of data gathered from 1988 till 2024 in order to measure the attitudes towards same sex marriage, age, and political party affiliation. Using logistic regression is appropriate for modeling binary outcomes like support for same-sex marriage. It allowed us to figure the estimation of how both variables: age (continuous) and political affiliation (categorical) are related to the probability of support. Overall, with the data and modeling approach it helps address the research question by analyzing how the demographic and political factors relate to supporting same sex marriage.

Results

Confusion Matrix		
Training		
Actual \ Predicted	1	0
Support_SSM	Count	
1	3262	1041
0	1979	1905

(1= supports same sex marriage, 0 = does not support)

Accuracy= 63.1%, Sensitivity= 75.8%, Specificity= 49.1%

Conclusion

By looking over the results, it seems that older adults are less likely to support same sex marriage, with Republicans showing the lowest probabilities overall. This is not a surprise since the Republican views and ideology tend to follow christianity which goes against same sex marriage. This shows consistency with GSS trends of generational and supporter gaps in social liberalization. Democrats exhibit nearly 9 fold higher odds than Republicans, highlighting ideology's role. These findings reveal ongoing cultural shifts, suggesting that targeted education for older conservatives could generate greater inclusivity, though broader societal factors such as media exposure can influence attitudes.

Report 2: Test of Independence - Association Between Gender and Views on Welfare Spending

Research Question and Context

The GSS variable NATFARE collects attitudes toward the amount of government spending on welfare. Respondents ask:

"We are faced with many problems in this country... Are we spending too much, too little, or about the right amount on welfare?"

Responses are categorized as: "too little," "about right," or "too much."

Using this given data set, this report determines whether there is a statistically significant association between gender (male or female) and views on government spending.

While debating about welfare attitudes in the U.S., shaped by inequality, poverty, and fiscal policy, the GSS data showed persistent gender gaps, where women tend to favor more generous social spending than men. This trend could possibly affect women and families the most due to disproportionate impacts of poverty. Although supporting increased welfare has been the factor depending on economic conditions, women consistently express more supportive views, showing significance on gender equity in policy design.

Statistical Method and Assumptions

A chi-square test of independence was conducted in JMP's contingency platform for the 2×3 table (gender $\times$ spending views). The test statistic follows a χ^2 distribution with $(r-1)(c-1) = 2$ df.

Assumptions:

(1) categorical variables

- (2) independent observations
- (3) expected cell frequencies ≥ 5 (all met, minimum 512)
- (4) random sample (GSS design ensures this).

Justification

This test evaluates whether the distribution of welfare views differs by gender beyond random variation, ideal for nominal GSS data. Cross tabulations allow direct comparison of proportions, answering the question by quantifying association strength via a contingency table.

Results

Contingency table (n = 62,000):

Gender \ View	Too Little	About Right	Too Much	Row Total
Male	12,450 (40%)	9,800 (32%)	8,750 (28%)	31,000
Female	15,300 (49%)	8,500 (27%)	7,200 (23%)	31,000
Column Total	27,750	18,300	15,950	62,000

We'll use this table to show all the statistics by hand:

- Rows: Gender = Male, Female
- Columns: View = Too Little, About Right, Too Much

So:

- Row totals:
 - $n_{1.} = 31,000$ (Male)
 - $n_{2.} = 31,000$ (Female)
- Column totals:
 - $n_{.1} = 27,750$ (Too Little)
 - $n_{.2} = 18,300$ (About Right)
 - $n_{.3} = 15,950$ (Too Much)
- Overall total: $N = 62,000$
- H_0 : Gender and welfare spending attitudes are independent.
- H_a : Gender and welfare spending attitudes are associated (not independent).

Expected Counts Under H_0 :

Under independence, the expected count in cell (row i, column j) is:

$$E_{ij} = \frac{(\text{Row total})(\text{Column total})}{N}$$

We compute E_{ij} for the 6 cells:

$$E_{11} = 13875, E_{12} = 9150, E_{13} = 7975, E_{21} = 13875, E_{22} = 9150, E_{23} = 7975$$

Expected Table:

Gender \ View	Too Little	About Right	Too Much	Row Total
Male	13,875	9,150	7,975	31,000
Female	13,875	9,150	7,975	31,000
Column Total	27,750	18,300	15,950	62,000

Chi-Square Test Stat: $\chi^2 = \sum_{i=1}^2 \sum_{j=1}^3 \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

We compute each cell:

$$\text{Cell (Male, Too Little)} = 146.4$$

$$\text{Cell (Male, About Right)} = 46.2$$

$$\text{Cell (Male, Too Much)} = 75.3$$

$$\text{Cell (Female, Too Little)} = 146.4$$

$$\text{Cell (Female, About Right)} = 46.2$$

$$\text{Cell (Female, About Right)} = 75.3$$

Test statistic: $\chi^2 \approx 146.4 + 46.2 + 75.3 + 146.4 + 46.2 + 75.3 = \mathbf{535.8}$

Degrees of Freedom = $df = (2-1)(3-1) = 1 \times 2 = \mathbf{2}$

Critical value for $\alpha = 0.05$: $\chi^2_{0.05,2} \approx \mathbf{5.99}$

Decision Using χ^2 Distribution

We compare our test statistic to the chi-square distribution with 2 degrees of freedom.

Since $535.8 > 5.99$, the p-value is clearly less than 0.05. Therefore, we reject H_0 and conclude that there is very strong evidence of an association between gender and welfare spending attitudes.

Interpretation in Words

Based on the observed percentages (using the original table):

For Men:

- Too Little: 40.2%
- About Right: 31.6%
- Too Much: 28.2%

For Women:

- Too Little: 49.4%
- About Right: 27.4%
- Too Much: 23.2%

We see that:

- A majority of women say “Too little” at about 49% compared to men at about 40%.
- A majority of men say “Too much” at about 28% compared to women at about 23%.
- A majority of men also say “About right” at about 32% compared to women at about 27%.

Conclusion:

Based on this result, the majority of women are highly likely to believe that the government spends too little on welfare, while the majority of men are highly likely to believe that the government either spends too much or about the right amount. This result means that men and women have different opinions and point-of-views, which is consistent with the gender gaps. The chi-square test confirms that the difference in these distributions is far greater than we would expect by random sampling variation alone.

Report 3: Kruskal-Wallis Test - Happiness Differences Across Income Levels

Research Question and Context

The GSS variable Happy asks:

“Taken all together, how would you say things are these days—very happy, pretty happy, or not too happy?”

Responses are coded ordinally:

1 = not too happy, 2 = pretty happy, 3 = very happy.

Income is measured by an ordinal version of RINCOME grouped into 10 brackets (1 = under \$10,000, ... , 10 = \$150,000+ annually). We ask:

Do median happiness levels differ across income categories?

Prior research suggests that happiness is the key in increasing income levels, though they can decrease at times regardless of how much happiness is produced. Testing this relationship helps predict the relationship between material resources and subjective well being.

Statistical Method and Assumptions

We apply the Kruskal–Wallis test, a nonparametric analogue of one way ANOVA, via JMP's nonparametric platform.

- Response: Happy (ordinal).
- Grouping factor: income bracket (10 ordered categories).

The test statistic H approximately follows a χ^2 distribution with $df = \text{number of groups} - 1 = 9$.

Assumptions:

1. The outcome is at least ordinal.
2. Observations are independent.
3. Group distributions have similar shapes so that differences are interpretable as shifts in central tendency (checked using histograms/boxplots).

Justification

Because happiness is ordinal and the income variable is grouped into broad brackets, the normality and equal-variance assumptions underlying ANOVA are questionable. Kruskal–Wallis instead compares rank distributions across groups and is more robust to skewness and unequal variances, making it well suited to these GSS variables.

Results

The Kruskal-Wallis test is highly significant:

- $H = 1,245.6$
- $df = 9$
- $p < 0.001$

Median happiness scores rise from about 1.8 in the lowest income bracket to about 2.6 in the highest bracket. The results using the Kruskal-Willist test show that lower income groups differ strongly from higher ones (lowest vs highest income: $z = 28.4$, $p < 0.001$).

After using JMP to create boxplots, an upward shift occurs in the distribution to show happiness by income bracket, where lower brackets are centered around “not too happy” to “pretty happy,” whereas the highest brackets cluster closer to “very happy.”

Conclusion

We find strong evidence that happiness varies by income level, with higher income associated with higher median happiness. However, the only setback is that not involving factors such as health, relationships, and job quality likely play an important additional role in shaping life satisfaction. This shows that although income levels play a role in happiness, personal lives that are not related to monetary factors also matter since happiness can vary either way.

Report 4: Measures of Rank Correlation - Education and Subjective Class Identification

Research Question and Context

GSS EDUC records years of education. CLASS captures subjective class and asks:

"If you were asked to use one of four names for your social class, which would you say you belong in: the lower class, the working class, the middle class, or the upper class?"

(ordinal: 1 = lower, 2 = working, 3 = middle, 4 = upper)

This report assesses:

Is there a positive monotonic correlation between education years and class identification?

GSS data links education to social mobility, with higher education often elevating perceived class amid U.S. stratification trends. This explores education's role in identity formation.

Statistical Method and Assumptions

Spearman's ρ and Kendall's τ (JMP's Multivariate platform) measure rank associations.

Assumptions:

(1) ordinal/continuous variables

(2) monotonic relationship

(3) independence.

Justification

These nonparametric metrics quantify strength/direction without a linearity assumption, fitting GSS ordinal class and continuous education.

Results

By using JMP, the results gave Spearman's $\rho = 0.215$ and Kendall's $\tau = 0.172$, where $p < 0.001$, which is a weak positive and about 62% concordant pairs.

Conclusion

Education shows a modest but statistically significant positive association with subjective social class. Individuals with higher degrees tend to have better chances of belonging to higher social classes due to the amount of schooling experience. This means that education is a key and symbol of social mobility. At the same time, the relatively small magnitude of the correlations implies that occupation, income, family background, and other factors also materially shape how people view their place by levels of social classes. This shows that individuals with higher education backgrounds are more likely to belong in the upper class with better resources.